

TITLE

Data Science Research Capstone – Time-Series Demand Forecasting Using ARIMA

Data Science Research Capstone & Whitepaper

Prepared

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Time-Series Forecasting with ARIMA

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Abstract

This study presents an end-to-end time-series forecasting approach for predicting daily demand using the ARIMA model. Historical demand data containing date, demand, marketing event, and holiday information was analyzed through data preprocessing and exploratory data analysis. The stationarity of the demand series was evaluated using the Augmented Dickey-Fuller test. The original and first-differenced series were found to be non-stationary, while second-order differencing achieved stationarity. An ARIMA(1,2,1) model was therefore developed using an 80% training and 20% testing split. The model achieved a Mean Absolute Percentage Error (MAPE) of 7.41% and a Root Mean Squared Error (RMSE) of 14.23 on the test dataset. A 30-observation future forecast was also generated. The results demonstrate that ARIMA can provide useful demand estimates for supporting inventory planning, resource allocation, operational planning, and data-driven decision-making.

Keywords

Keywords: Time-Series Forecasting, ARIMA, Demand Forecasting, Stationarity, ADF Test, Data Science, Predictive Analytics

Research Question

Can historical daily demand data be used to accurately forecast future demand using the ARIMA time-series forecasting model?

Research Objectives

1. To analyze historical daily demand patterns.
2. To identify trends and patterns in the demand data.
3. To test the stationarity of the time series.
4. To apply appropriate differencing to achieve stationarity.
5. To develop an ARIMA forecasting model.
6. To evaluate the model using MAPE and RMSE.
7. To generate future demand forecasts.
8. To understand the practical and business implications of demand forecasting.

Research Hypothesis

Null Hypothesis (H_0):

Historical demand data cannot provide reliable information for forecasting future demand.

Alternative Hypothesis (H_1):

Historical demand data can be used to generate reliable forecasts of future demand using the ARIMA model.

Dataset Description

The dataset used in this study is a daily demand time-series dataset. It contains historical observations recorded by date along with demand-related information. The dataset includes the date, demand, marketing event, and holiday variables. The demand variable is considered the primary target variable for the forecasting task, while marketing event and holiday information provide additional contextual information about demand patterns.

Dataset Variables

Variable	Description	Role
Date	Date associated with each daily observation	Time index
Demand	Recorded demand value for the corresponding date	Target variable
Marketing Event	Indicator showing whether a marketing event occurred	Contextual variable
Holiday	Indicator showing whether the date was a holiday	Contextual variable

Data preparation

Before performing the forecasting analysis, the date variable was converted into a proper datetime format. The observations were then sorted chronologically according to the date. The date column was subsequently set as the index of the dataset so that the data could be treated as a time series. This preparation ensured that the observations were arranged in the correct temporal order for subsequent analysis and forecasting.

Forecasting Variable

The primary target variable selected for forecasting is **demand**. The objective of the forecasting process is to use historical demand observations to estimate future demand.

values. Since demand is observed sequentially over time, it is treated as a time-series variable.

Methodology

The study follows an end-to-end time-series forecasting workflow. The process begins with loading and preparing the historical demand dataset, followed by exploratory data analysis to understand the underlying patterns. The stationarity of the demand series is then examined using the Augmented Dickey-Fuller (ADF) test. When the original series is found to be non-stationary, differencing is applied to obtain a stationary series. An ARIMA model is subsequently developed and evaluated using appropriate forecasting performance metrics. Finally, the trained model is used to generate future demand forecasts.

Stationarity Testing

Stationarity is an important requirement for ARIMA-based time-series forecasting. The Augmented Dickey-Fuller (ADF) test was used to determine whether the demand series was stationary. A p-value below 0.05 indicates rejection of the null hypothesis of a unit root and suggests that the series is stationary. Since the original demand series was non-stationary, differencing was applied progressively until stationarity was achieved.

ADF Test Results

Series	ADF Statistic	p-value	Conclusion
Original Demand	-0.903	~0.787	Non-stationary
First Difference	-2.234	0.194	Non-stationary
Second Difference	-15.187	6.085×10^{-9}	Stationary

Interpretation

The ADF test results show that the original demand series was non-stationary because its p-value was greater than 0.05. After applying first-order differencing, the series remained non-stationary, with a p-value greater than 0.05. After applying second-order differencing, the p-value became significantly lower than 0.05, indicating that the series had achieved stationarity. Therefore, second-order differencing was selected for the ARIMA model, resulting in $d = 2$.

Model Development and Evaluation

ARIMA Model Development

Based on the stationarity analysis, an ARIMA model was developed for forecasting daily demand. Since second-order differencing was required to achieve stationarity, the differencing parameter was selected as $d = 2$. The selected model was ARIMA(1,2,1), where $p = 1$ represents the autoregressive component, $d = 2$ represents second-order differencing, and $q = 1$ represents the moving-average component.

Train-Test Split

The historical demand data was divided into training and testing sets. Approximately 80% of the observations were used to train the ARIMA model, while the remaining 20% were reserved for testing. The test data was not used during model training and was therefore used to evaluate the model's forecasting performance on unseen observations.

Model Evaluation

Evaluation Metric Result

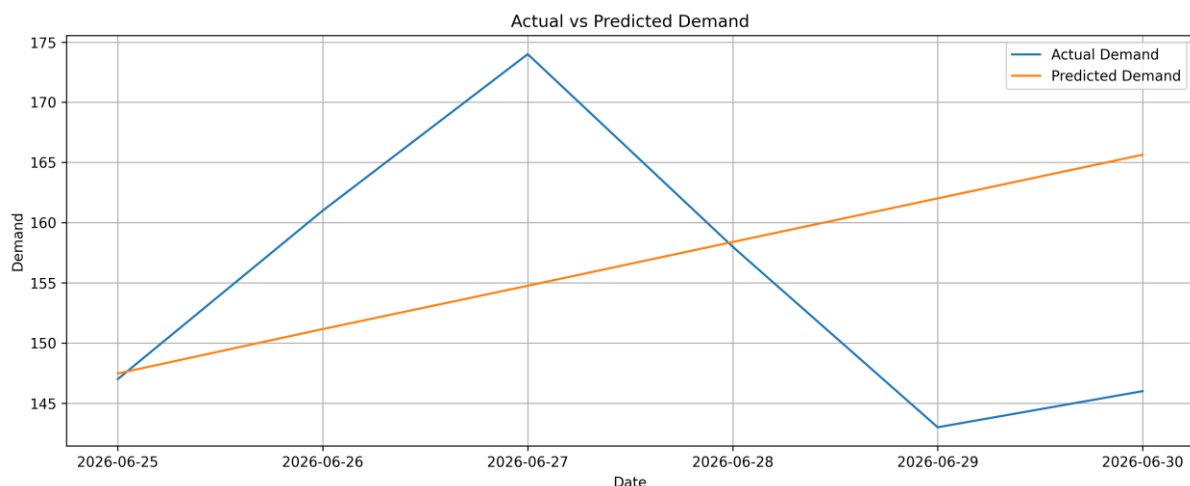
MAPE	7.41%
RMSE	14.23

Interpretations of Result

The ARIMA(1,2,1) model achieved a MAPE of 7.41% and an RMSE of 14.23 on the test dataset. The MAPE indicates that the average forecasting error was approximately 7.41% relative to the actual demand values. The RMSE represents the magnitude of prediction errors in the original demand units. Overall, the evaluation results indicate that the model provides reasonably accurate predictions for the analyzed demand series.

Actual vs Predicted Demand Graph

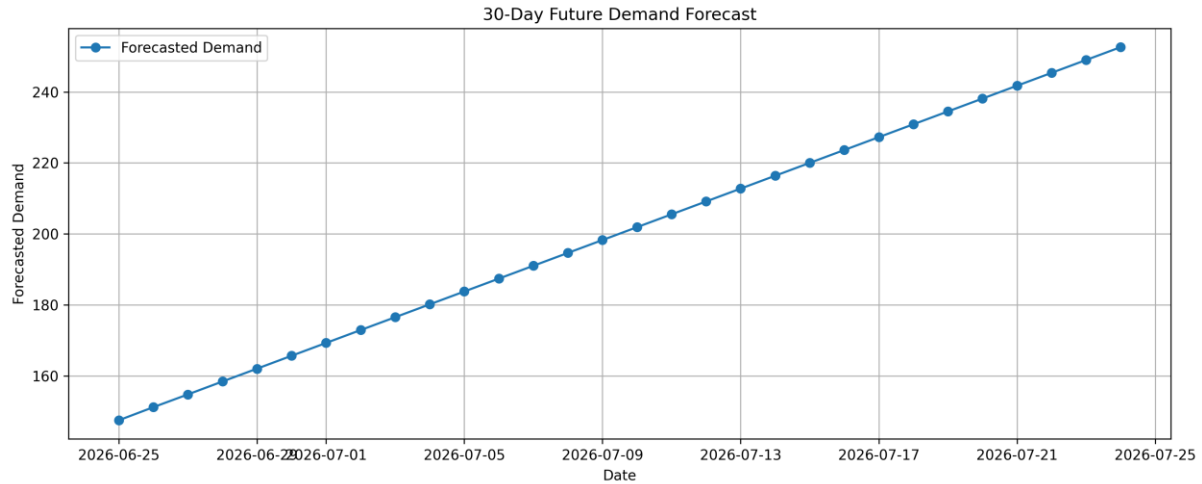
Figure 1. Actual vs Predicted Demand During the Test Period



Future Demand Forecasting

After evaluating the ARIMA(1,2,1) model on the test dataset, the fitted model was used to generate future demand forecasts. A 30-observation forecasting horizon was selected to estimate the expected demand for the upcoming period. The generated forecasts provide an estimate of future demand based on the historical patterns captured by the ARIMA model.

Figure 2. Thirty-Day Future Demand Forecast



Forecasted demand values

Date	Forecasted demand
2026-06-25	147.469242
2026-06-26	151.159280
2026-06-28	154.757466
2026-06-29	158.393863
2026-06-30	162.014364
2026-07-01	165.641478
2026-07-02	169.265841
2026-07-03	172.891348
2026-07-04	176.516379
2026-07-05	180.141608
2026-07-06	183.766755
2026-07-07	187.391936
2026-07-08	191.017103
2026-07-09	198.267446
2026-07-10	201.892618
2026-07-11	205.517789
2026-07-12	209.142960
2026-07-13	212.768131
2026-07-14	216.393302
2026-07-15	220.018473
2026-07-16	223.643644
2026-07-17	227.268815
2026-07-18	230.893986
2026-07-20	238.144328
2026-07-21	241.769499
2026-07-22	245.394670

2026-07-23	249.019841
2026-07-24	252.645012

Business Implications

The developed ARIMA-based demand forecasting model can support data-driven decision-making by providing estimates of future demand. Accurate demand forecasts can help organizations improve inventory planning, resource allocation, workforce scheduling, and operational planning.

The forecasting results can be used to anticipate periods of higher or lower demand and take appropriate actions in advance. For example, organizations can adjust inventory levels, allocate resources efficiently, and plan operational activities according to expected demand.

The model can also support marketing and promotional planning. Historical demand patterns combined with information about marketing events and holidays can help decision-makers understand changes in demand and improve future planning strategies.

The ARIMA(1,2,1) model achieved a MAPE of 7.41% and an RMSE of 14.23 on the test dataset, indicating reasonably accurate forecasting performance for the analyzed demand series. However, forecasts should be periodically updated as new observations become available, since changes in market conditions, customer behavior, seasonal patterns, or external events may affect future demand.

Key Business Benefits

1. **Improved Demand Planning** – Future demand estimates can help organizations prepare for expected changes in demand.
2. **Inventory Optimization** – Forecasts can support better inventory-level decisions and reduce the risk of overstocking or stock shortages.
3. **Resource Allocation** – Organizations can allocate employees, equipment, and other resources according to expected demand.
4. **Operational Efficiency** – Advance knowledge of expected demand can improve operational planning and reduce unnecessary costs.
5. **Marketing Planning** – Demand patterns can help organizations evaluate and plan marketing activities and promotions.
6. **Data-Driven Decision Making** – Forecasting provides quantitative evidence that can support business decisions rather than relying only on assumptions.

Limitations

The forecasting model has some limitations. ARIMA primarily relies on historical patterns and may not fully capture sudden changes caused by unexpected events, major market changes, or other external factors. The model performance may also vary when the underlying demand pattern changes significantly. Additionally, the current model can be further improved by incorporating relevant external variables and comparing ARIMA with other forecasting approaches.

Research Findings

The analysis demonstrates that the daily demand series initially exhibited non-stationary behavior. The Augmented Dickey-Fuller test showed that the original series and first-differenced series were non-stationary, while the second-differenced series achieved stationarity. Therefore, a differencing order of $d = 2$ was selected for the ARIMA model.

The ARIMA(1,2,1) model was trained using approximately 80% of the available observations and evaluated on the remaining 20% test observations. The model achieved a MAPE of 7.41% and an RMSE of 14.23, indicating reasonably accurate predictive performance for the analyzed dataset.

The 30-observation future forecast provides estimated demand values for the upcoming period. These forecasts can support demand planning, inventory management, resource allocation, and other operational decisions.

Conclusion

This study developed an end-to-end time-series forecasting pipeline for predicting daily demand using the ARIMA methodology. The analysis included data preprocessing, exploratory data analysis, stationarity testing, differencing, model development, prediction, and evaluation.

The ADF test established that second-order differencing was required to achieve stationarity. Based on this analysis, an ARIMA(1,2,1) model was developed. The model demonstrated reasonably accurate forecasting performance on the test dataset, with a MAPE of 7.41% and an RMSE of 14.23.

The resulting future forecasts demonstrate the practical potential of time-series analysis for supporting data-driven planning and decision-making. Future work could explore seasonal ARIMA (SARIMA), machine-learning models, or additional external variables to potentially improve forecasting performance.

Literatures Background

Time-series forecasting is an important area of data science used to predict future values based on historical observations. It has applications in demand forecasting, sales prediction, inventory management, finance, energy consumption, and resource planning.

Traditional statistical forecasting methods such as Autoregressive Integrated Moving Average (ARIMA) are widely used because they can model temporal dependencies and trends in time-series data. ARIMA combines autoregression, differencing, and moving-average components to model non-stationary time-series data.

Stationarity is an important consideration when developing ARIMA models. A stationary series has statistical properties that remain relatively stable over time. The Augmented Dickey-Fuller (ADF) test is commonly used to assess stationarity. When a series is non-stationary, differencing can be applied to transform it into a stationary series.

Previous research has demonstrated the usefulness of statistical and machine-learning approaches for demand forecasting. However, model performance depends on the characteristics of the dataset, forecasting horizon, seasonal patterns, and external factors. Therefore, evaluating forecasting models using appropriate metrics such as MAPE and RMSE is important.

Based on this background, the present study investigates an ARIMA-based approach for daily demand forecasting and evaluates its performance using historical demand observations.

Research Gap

Although time-series forecasting methods are widely used, forecasting performance can vary depending on the characteristics of the demand series. Non-stationarity, changing demand patterns, marketing events, holidays, and other external factors can influence forecasting accuracy. Therefore, there is a need to systematically analyze the demand series, establish stationarity, develop an appropriate forecasting model, and evaluate its performance on unseen data. This study addresses this requirement through an end-to-end ARIMA forecasting pipeline.

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